

Ex.No.7 Use AFLogical OSE to extract data from an Android device

Description

AFLogical OSE (Open Source Edition) is a tool used to extract data from Android devices. It's part of the Open Source Android Forensics project and is often used for logical extraction, which means pulling data like contacts, messages, and call logs without directly accessing the file system. Here's a step-by-step guide to using AFLogical OSE to extract data from an Android device.

Step 1: Prepare Your Environment

1. Download AFLogical OSE:

- Visit the [AFLogical OSE GitHub page](#) and download the latest release. Alternatively, you can clone the repository using Git.

2. Install Java (if not already installed):

- AFLogical OSE requires Java to run. Make sure you have Java installed on your machine. You can download it from the [official Java website](#).

3. Install Android Debug Bridge (ADB):

- ADB is a command-line tool that allows communication with Android devices. Download it from the [Android Developer's website](#) and install it on your system.
- Add ADB to your system's PATH environment variable to easily run ADB commands from the terminal or command prompt.

4. Enable USB Debugging on the Android Device:

- On the Android device, go to Settings > About Phone and tap Build Number seven times to enable Developer Options.
- Go to Settings > Developer Options and enable USB Debugging.

Step 2: Connect the Android Device to Your Computer

1. Connect via USB:

- Use a USB cable to connect the Android device to your computer.

2. Verify ADB Connection:

- Open Command Prompt (Windows) or Terminal (macOS/Linux) and run:

bash

adb devices

- This command lists connected devices. If your device is listed, you are ready to proceed. If not, ensure that USB Debugging is enabled and that the correct drivers are installed.

Step 3: Extract Data Using AFLogical OSE

1. Run AFLogical OSE:

- Navigate to the directory where you have downloaded or extracted AFLogical OSE.

2. Push the APK to the Android Device:

- Run the following command to install the AFLogical OSE APK on the device:

bash

adb install aflogical.apk

- This installs the AFLogical app on the device, which will be used to collect data.

3. Launch the AFLogical OSE App on the Device:

- On the Android device, open the AFLogical app.
- You'll see options to extract various types of data (e.g., contacts, SMS, MMS, call logs). Select the data types you want to extract.

4. Start Data Extraction:

- Once you've selected the data types, the app will start the extraction process.
- Data is stored in .csv files on the device's storage, typically in a directory named aflogical.

Step 4: Transfer Extracted Data to Your Computer

1. Use ADB to Pull Data:

- After extraction, you need to transfer the data files from the Android device to your computer. Run:

bash

adb pull /sdcard/aflogical /path/to/destination

- Replace /path/to/destination with the directory path where you want to save the extracted data on your computer.

2. Verify the Data:

- Navigate to the destination directory and check the .csv files to ensure all required data has been extracted.

Step 5: Analyze the Extracted Data

1. Open the CSV Files:

- Use Excel, Google Sheets, or any text editor to open and analyze the .csv files. These files contain the extracted data like contacts, SMS, MMS, and call logs.

2. Review and Document:

- Carefully review the data for any relevant information. Document your findings and consider exporting the data into a report format for further analysis.

Step 6: Clean Up

1. Uninstall AFLogical OSE:

- Once data extraction is complete, you can uninstall the AFLogical OSE app from the Android device by running:

bash

adb uninstall com.viaforensics.android.aflogical

2. Disconnect the Device:

- Safely disconnect the Android device from your computer.

By following these steps, you can use AFLogical OSE to effectively extract and analyze data from an Android device. This process is valuable for forensic investigations, especially when dealing with logical data that doesn't require root access to the device.

OUTPUTS:

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\saty>c:\platform-tools
'C:\platform-tools' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\saty>dir C:\platform-tools
Volume in drive C is Windows
Volume Serial Number is 106D-CD90

Directory of C:\platform-tools

20-09-2026 16:58 <DIR> .
20-09-2026 16:45      8,273,560 adb.exe
20-09-2026 16:45     108,184 AdbWinApi.dll
20-09-2026 16:45      73,368 AdbWinUsbApi.dll
20-09-2026 16:45     451,224 etc1tool.exe
20-09-2026 16:45    2,429,080 fastboot.exe
20-09-2026 16:45      52,888 hprof-conv.exe
20-09-2026 16:45     243,816 libwinpthread-1.dll
20-09-2026 16:45     473,752 make_f2fs.exe
20-09-2026 16:45     473,752 make_f2fs_casefold.exe
20-09-2026 16:45       1,157 mke2fs.conf
20-09-2026 16:45     763,544 mke2fs.exe
20-09-2026 16:45    1,154,131 NOTICE.txt
20-09-2026 16:45         38 source.properties
20-09-2026 16:45    3,031,704 sqlite3.exe
                14 File(s)      17,529,398 bytes
                1 Dir(s)      239,791,525,888 bytes free

C:\Users\saty>
```

```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\saty>adb pair 10.1.4.145:37711
Enter pairing code: 856522
Successfully paired to 10.1.4.145:37711 [guid=adb-R2CX50T016B-6uhg6i]

C:\Users\saty>adb devices
List of devices attached
adb-R2CX50T016B-6uhg6i._adb-tls-connect._tcp    device

C:\Users\saty>mkdir C:\forensics-output
A subdirectory or file C:\forensics-output already exists.

C:\Users\saty>cd C:\forensics-output

C:\forensics-output>adb shell content query --uri content://call_log/calls > calllog.txt
C:\forensics-output>callog.txt
C:\forensics-output>adb shell content query --uri content://contacts/phones > contacts.txt
adb shell content query --uri content://sms > sms.txt
C:\forensics-output>
C:\forensics-output>contacts.txt
C:\forensics-output>sms.txt
'sms.txt' is not recognized as an internal or external command,
operable program or batch file.
C:\forensics-output>adb shell content query --uri content://sms > sms.txt
C:\forensics-output>adb shell content query --uri content://contacts/phones > contacts.txt
C:\forensics-output>contacts.txt
C:\forensics-output>sms.txt
C:\forensics-output>adb disconnect
'adb' is not recognized as an internal or external command,
operable program or batch file.
C:\forensics-output>
```

Result: The experiment demonstrated logical data extraction from an Android device without requiring root access. Since the original AFLogical OSE application is no longer available for download (its official release has been discontinued), the same objective was achieved using ADB's content query command to directly query the device's content providers for call logs, contacts, and SMS messages. Data was extracted successfully over a wireless ADB connection and transferred to the host computer as structured text output, which was then parsed and organized into readable tables for analysis. This confirms that logical extraction of user data — including call history, contact details, and messages — can be performed effectively through standard ADB commands, providing a viable modern alternative to dedicated forensic acquisition tools when such tools are unavailable.